

EDA Report – Automobile Dataset

Introduction

Objective of the Analysis

The objective of this exploratory data analysis (EDA) is to investigate the automobile dataset, identify patterns and relationship within the data, and gain insights into factors that influence automobile prices. The analysis includes data cleaning, preprocessing, visualisation and interpretation of key trends.

Dataset Description

The automobile dataset contains information about various vehicles, including manufacturer details, engine specifications, fuel type, horsepower, dimensions and price. These attributes provide valuable information for understanding the characteristics that contribute to vehicle pricing and market segmentation.

Data Cleaning

Prior to analysis, several data cleaning steps were performed to improve the quality and reliability of the dataset.

Removal of duplicate Rows

The dataset was examined for duplicate records using the `duplicated()` function. Any duplicate rows identified were removed to ensure that each observation represented a unique vehicle.

Removal of Missing Values

Missing values can affect statistical analysis and visualisations. Rows containing missing values were identified and removed using the `dropna()` function. This ensured that the analysis was conducted using complete and reliable data.

Data Type Conversion

Several columns containing numerical information were initially stored as text. These columns were converted to appropriate numeric data types using `pd.to_numeric()`. e.g. price, Horsepower, Peak RPM, Bore, Stroke, Normalized Losses. Converting these variables enabled accurate calculations, correlations and visualisations.

Which are the 5 most expensive cars?

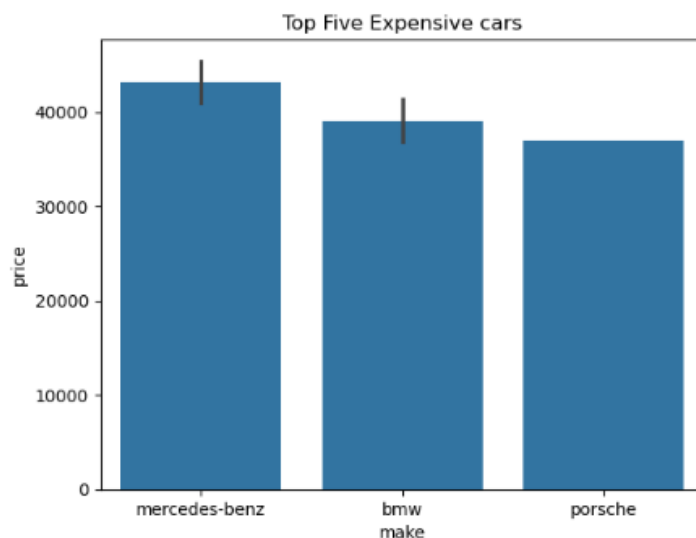
How do the most expensive and cheapest cars compare? Exploring the most expensive cars highlights if some movie cars are worth the money spent on them based on their fuel economy (mpg or miles per gallon)

The five most expensive automobiles were identified and compared with the five least expensive vehicles based on fuel economy measurements. The analysis indicated that high prices cars generally Offered lower fuel efficiency due to larger engines and performance-oriented designs. whereas cheaper cars tended to provide better mileage and economical operation.

```
[28]: top_five_expensive = automobiles_df.sort_values(by='price', ascending=False).head(5)
print(top_five_expensive[['make', 'price', 'wheel-base', 'length', 'width', 'height', 'curb-weight', 'engine-size', 'compression-ratio', 'city-mpg', 'highway-mpg']])
```

	make	price	wheel-base	length	width	height	curb-weight	engine-size	compression-ratio	city-mpg	highway-mpg
74	mercedes-benz	45400	112	199	72	55	3715	304	8	14	16
16	bmw	41315	103	193	67	53	3380	209	8	16	22
73	mercedes-benz	40960	120	208	71	56	3900	308	8	14	16
128	porsche	37028	89	168	65	51	2800	194	9	17	25
17	bmw	36880	110	197	70	56	3505	209	8	15	20

```
[63]: sns.barplot(data=top_five_expensive, x='make', y='price')
plt.title('Top Five Expensive cars')
plt.show()
```

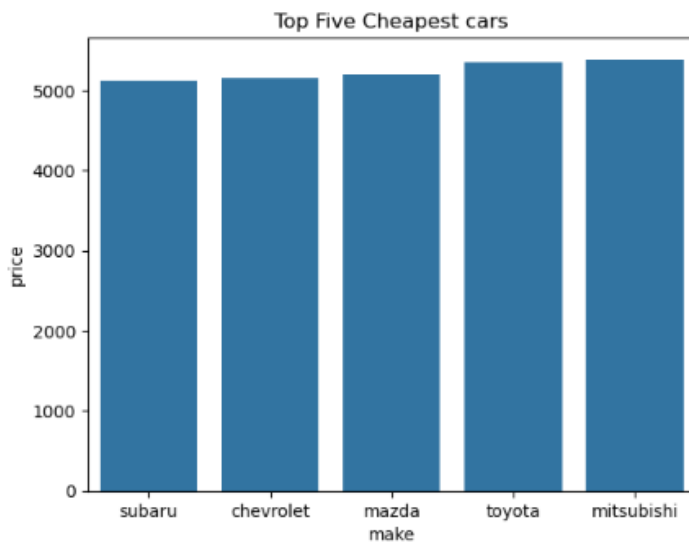


```
[29]: top_five_cheapest = automobiles_df.sort_values(by='price', ascending=True).head(5)
print(top_five_cheapest[['make', 'price', 'wheel-base', 'length', 'width', 'height', 'curb-weight', 'engine-size', 'compression-ratio', 'city-mpg', 'highway-mpg']])
```

	make	price	wheel-base	length	width	height	curb-weight	\
138	subaru	5118	93	156	63	53	2050	
18	chevrolet	5151	88	141	60	53	1488	
50	mazda	5195	93	159	64	54	1890	
150	toyota	5348	95	158	63	54	1985	
76	mitsubishi	5389	93	157	64	50	1918	

	engine-size	compression-ratio	city-mpg	highway-mpg
138	97	9	31	36
18	61	9	47	53
50	91	9	30	31
150	92	9	35	39
76	92	9	37	41

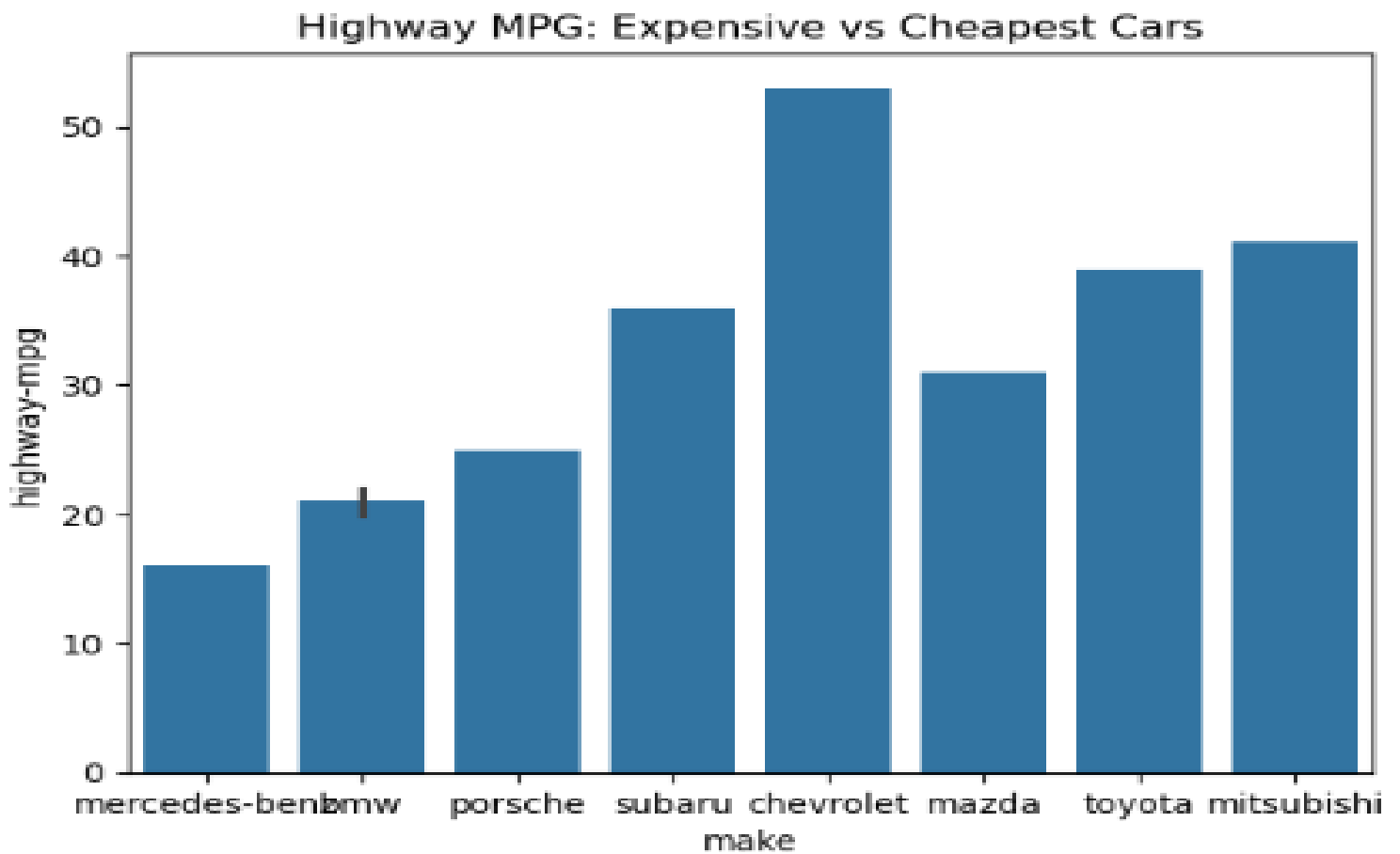
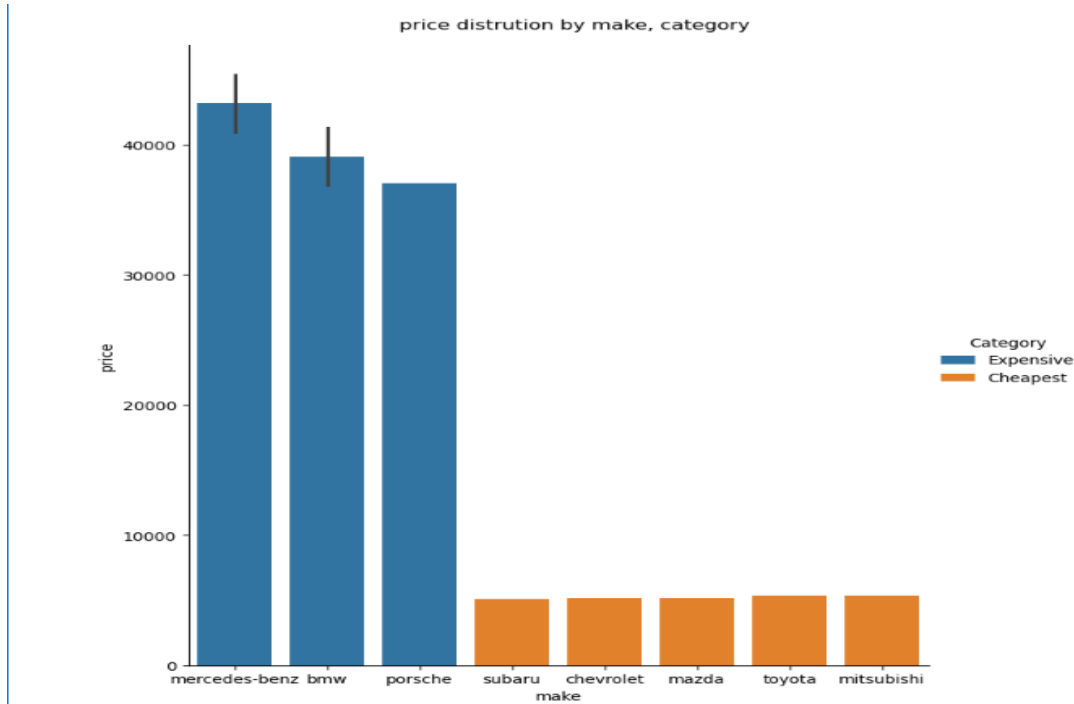
```
[62]: sns.barplot(data=top_five_cheapest, x='make', y='price')
plt.title('Top Five Cheapest cars')
plt.show()
```

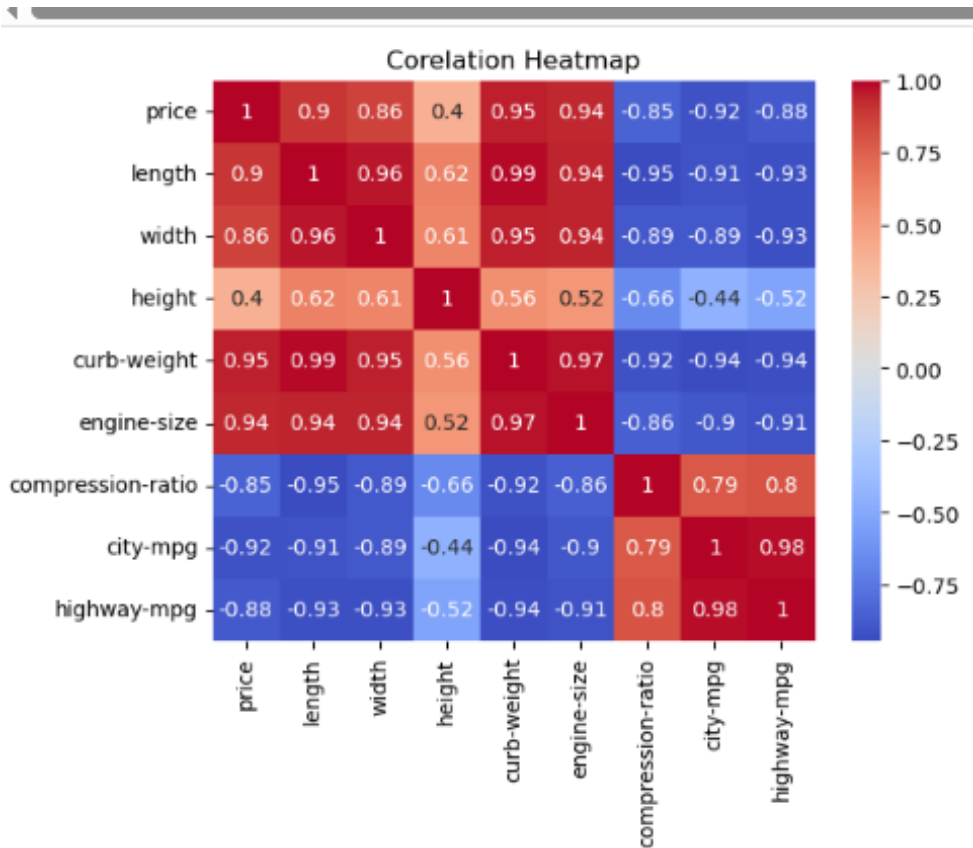


```
[30]: comarison_df = pd.concat([top_five_expensive.assign(Category='Expensive'),
                                top_five_cheapest.assign(Category='Cheapest')
                                ])
print(comarison_df[['make', 'price', 'length', 'width', 'height', 'curb-weight', 'engine-size', 'compression-ratio', 'city-mpg', 'highway-mpg']])
```

	make	price	length	width	height	curb-weight	engine-size	\
74	mercedes-benz	45400	199	72	55	3715	304	
16	bmw	41315	193	67	53	3380	209	
73	mercedes-benz	40960	208	71	56	3900	308	
128	porsche	37028	168	65	51	2800	194	
17	bmw	36880	197	70	56	3505	209	
138	subaru	5118	156	63	53	2050	97	
18	chevrolet	5151	141	60	53	1488	61	
50	mazda	5195	159	64	54	1890	91	
150	toyota	5348	158	63	54	1985	92	
76	mitsubishi	5389	157	64	50	1918	92	

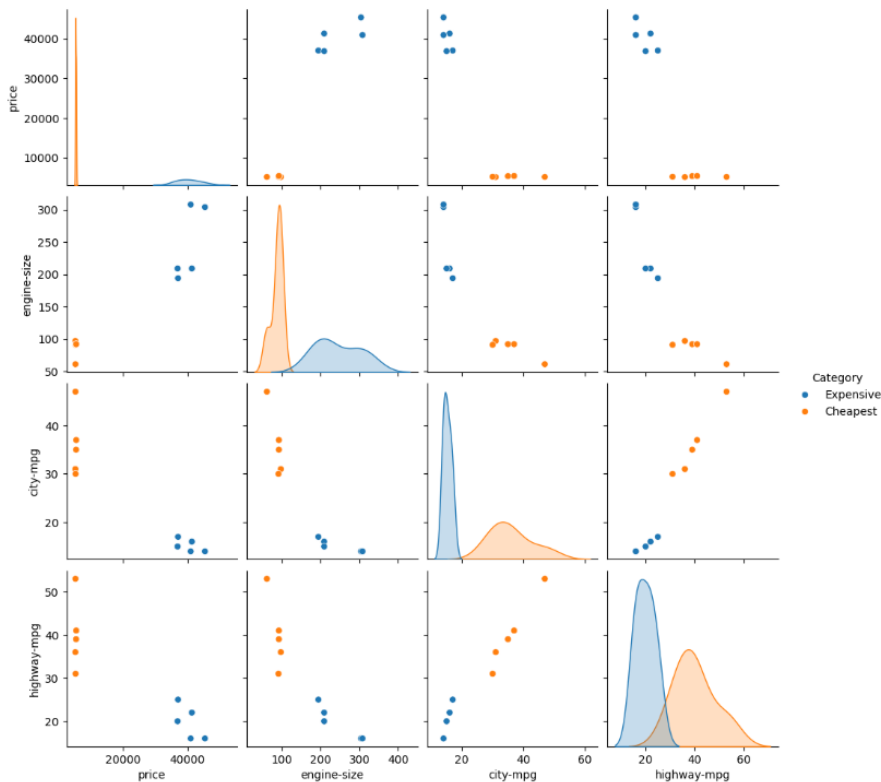
	compression-ratio	city-mpg	highway-mpg
74	8	14	16
16	8	16	22
73	8	14	16
128	9	17	25
17	8	15	20
138	9	31	36
18	9	47	53
50	9	30	31
150	9	35	39
76	9	37	41





Pair Plot - compare multiple variables

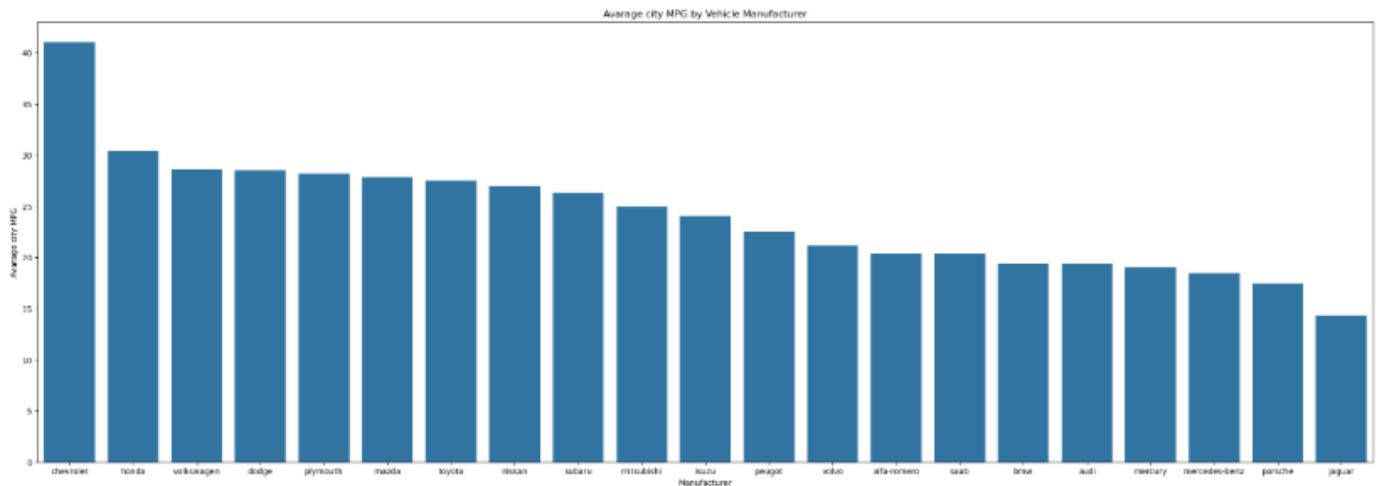
```
[38]: sns.pairplot(comarison_df, vars=['price', 'engine-size', 'city-mpg', 'highway-mpg'], hue='Category')
plt.show()
```



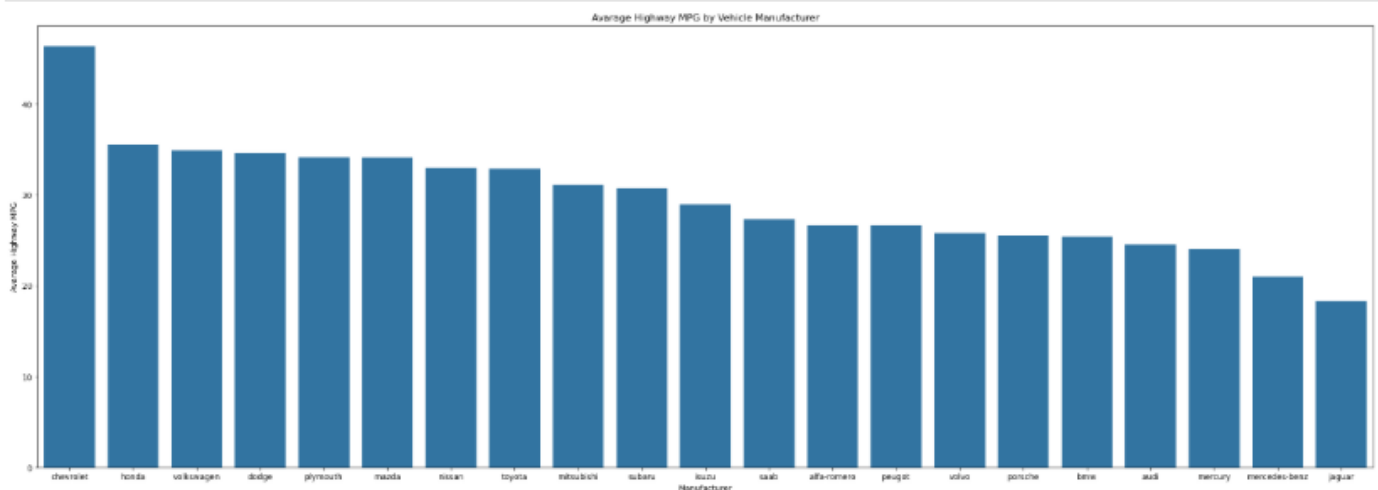
Which manufacturer builds the most fuel-efficient vehicles?

Compare the average mpg for each vehicle manufacturer's vehicles and create a bar plot

```
[40]: avg_city_mpg_df = avg_city_mpg.reset_index()
plt.figure(figsize=(30,10))
sns.barplot(data=avg_city_mpg_df , x='make', y='city-mpg' )
plt.title('Average city MPG by Vehicle Manufacturer')
plt.xlabel('Manufacturer')
plt.ylabel('Average city MPG')
plt.show()
```



```
[41]: avg_highway_mpg_df = avg_highway_mpg.reset_index()
plt.figure(figsize=(30,10))
sns.barplot(data=avg_highway_mpg_df , x='make', y='highway-mpg' )
plt.title('Average Highway MPG by Vehicle Manufacturer')
plt.xlabel('Manufacturer')
plt.ylabel('Average Highway MPG')
plt.show()
```



Which vehicles have the largest engine capacity.

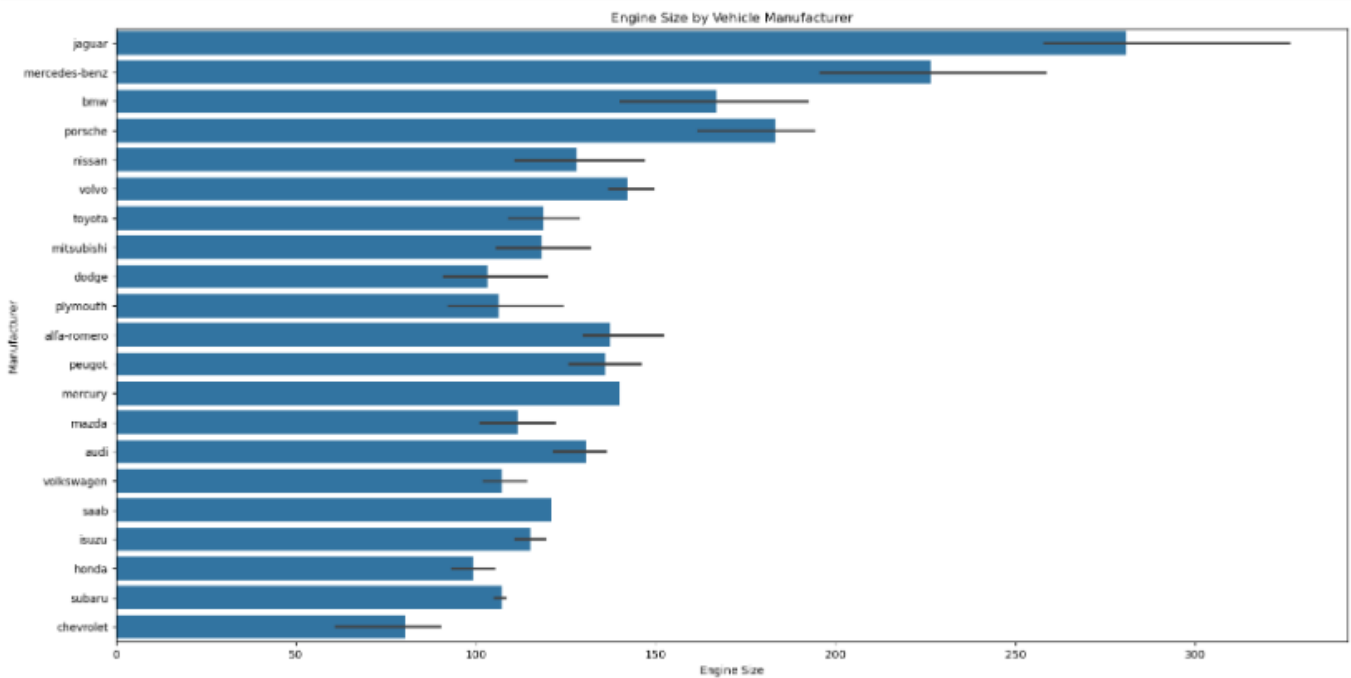
Sort the dataframe based on the engine-size column.

```
[42]: # Code here
largest_engine_size = automobiles_df.sort_values(by='engine-size', ascending=False)
print(largest_engine_size[['make', 'engine-size']])
```

```
   make  engine-size
49  jaguar         326
73 mercedes-benz    308
74 mercedes-benz    304
47  jaguar         258
48  jaguar         258
..    ...         ...
25  dodge          90
121 plymouth        90
118 plymouth        90
32  honda          79
18  chevrolet       61
```

[193 rows x 2 columns]

```
[43]: plt.figure(figsize=(20,10))
sns.barplot(data=largest_engine_size, x='engine-size', y='make')
plt.title('Engine Size by Vehicle Manufacturer')
plt.xlabel('Engine Size')
plt.ylabel('Manufacturer')
plt.show()
```



Which vehicle manufacturer has the most car models in the dataset

Which vehicle manufacturer has the most car models in the dataset

[44]: [# Code here](#)

```
manufacturer_count = automobiles_df['make'].value_counts()
print(manufacturer_count)
```

```
make
toyota      32
nissan      18
honda       13
mitsubishi  13
mazda       12
subaru       12
volkswagen  12
peugot      11
volvo       11
bmw         8
dodge       8
mercedes-benz 8
plymouth    7
audi        6
saab        6
porsche     4
alfa-romero 3
chevrolet   3
jaguar      3
isuzu       2
mercury     1
Name: count, dtype: int64
```

[45]:

```
print(manufacturer_count.idxmax())
print(manufacturer_count.max())
```

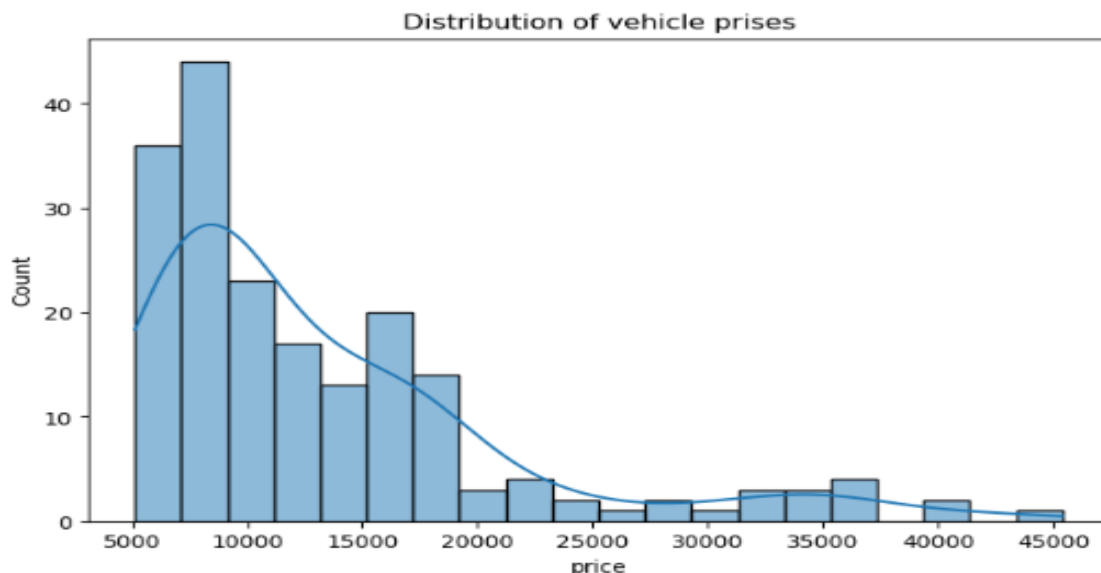
```
toyota
32
```

Exploratory Data Analysis

Price Distribution

#Visualisations

```
j: # Distribution of vehicle prises
plt.figure(figsize=(8,5))
sns.histplot(automobiles_df["price"], bins=20, kde=True)
plt.title("Distribution of vehicle prises")
plt.show()
```



The histogram shows that vehicle price is highly right-skewed (positively skewed). Most vehicles are concentrated in the lower price range (5000 to 15000), indicating that the dataset mainly consists of affordable or budget vehicles.

As the price increases the frequency of vehicles decreases sharply. There are relatively few mid-range vehicles (15000 to 25000) and only a small number of high-prices vehicles above 30000, with the distribution stretching up to around 45,000 forming a long right tail.

The KDE curves confirm this pattern showing a peak in the lower price range and a gradual decline as prices rise. This suggests the presence of outlier or luxury vehicles in the dataset,

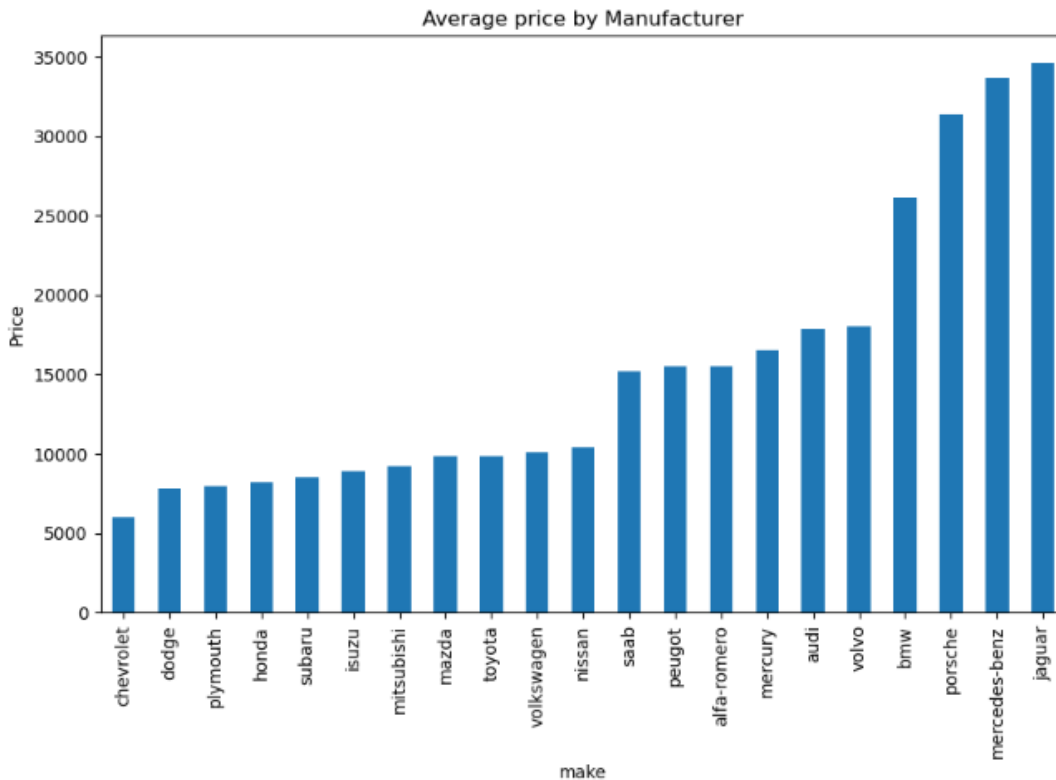
Key Insights:

- 1) Majority of cars are low-priced (5k – 15k range)
- 2) Strong positive skew in price distribution
- 3) Presence of high-value outliers (30k – 45k range)
- 4) Data is not normally distributed, so transformations may be useful for modelling

Average price by Manufacturer

```
[51]: # Average Price by make
avg_price = automobiles_df.groupby('make') ['price'].mean().sort_values()

plt.figure(figsize=(10,6))
avg_price.plot(kind='bar')
plt.title('Average price by Manufacturer')
plt.ylabel('Price')
plt.show()
```



EDA Summary – Average Price by Manufacturer

The bar chart compares the average vehicle price across different manufacturers. It reveals significant variation in pricing, highlighting the distinction between economy and luxury vehicle brands.

Chevrolet has the lowest average vehicle price at approximately \$6,000, followed by manufacturers such as Dodge, Plymouth, Honda, Subaru, Isuzu, Mitsubishi, Mazda, Toyota, Volkswagen and Nissan, whose average prices generally range between \$8,000 and \$11,000. These brands represent the more affordable segment of the market.

Manufacturers such as Saab, Peugeot, Alfa-Romeo, Mercury, Audi, and Volvo occupy the mid-range category, with average prices between \$15,000 and \$18,000.

The highest average prices are observed for luxury brands, with BMW, Porsche, Mercedes Benz, and Jaguar commanding significantly higher values. Jaguar has the highest average vehicle price at approximately \$35000 followed closely by Mercedes-Benz and Porsche.

Key Insights

- 1) Average vehicle prices vary substantially between manufacturers
- 2) Economy brands (Chevrolet, Dodge, Honda, Toyota, Nissan) have the lowest average prices.
- 3) Premium and luxury brands (BMW, Porsche, Mercedes-Benz, Jaguar) have significantly higher average prices.
- 4) Jaguar appears to be the most expensive manufacturer on average in this dataset
- 5) Manufacturer brand has a strong influence on vehicle pricing and could be an important predictor in price modelling.

The analysis indicates a clear relationship between manufacturer and vehicle price. Luxury manufacturers consistently command higher average prices than economy brands, suggesting that brand reputation, vehicle features and market positioning play a major role determining automobile prices. This variable is likely to be valuable for predictive modelling and market segmentation analyses

Horsepower vs Price

The scatter plot illustrates the relationship between horsepower and vehicle price. A clear positive correlation can be observed, indicating that vehicles with higher horsepower generally tend to have higher prices.

Most vehicles in the dataset are concentrated between 60 to 120 horsepower, with price ranging from \$5000 to 18000. As horsepower increases beyond this range, vehicle price tends to rise significantly, suggesting that engine performance is an important factor influencing automobile value

Key Insights

- 1) There is a strong positive relationship between horsepower and vehicle price.
- 2) Vehicles with lower horsepower (50-120 HP) are generally less expensive
- 3) Higher-horsepower vehicle tends to command higher market prices.
- 4) Several luxury vehicle outliers appear in the upper-right region of the plot with both high horsepower and high prices
- 5) Horsepower is likely to be an important predictor variable for price estimation models.

```
[53]: #Horsepower vs Price
```

```
plt.figure(figsize=(8,6))
sns.scatterplot(data=automobiles_df, x='horsepower', y='price')
plt.title('Horsepower vs Price')
plt.show()
```

